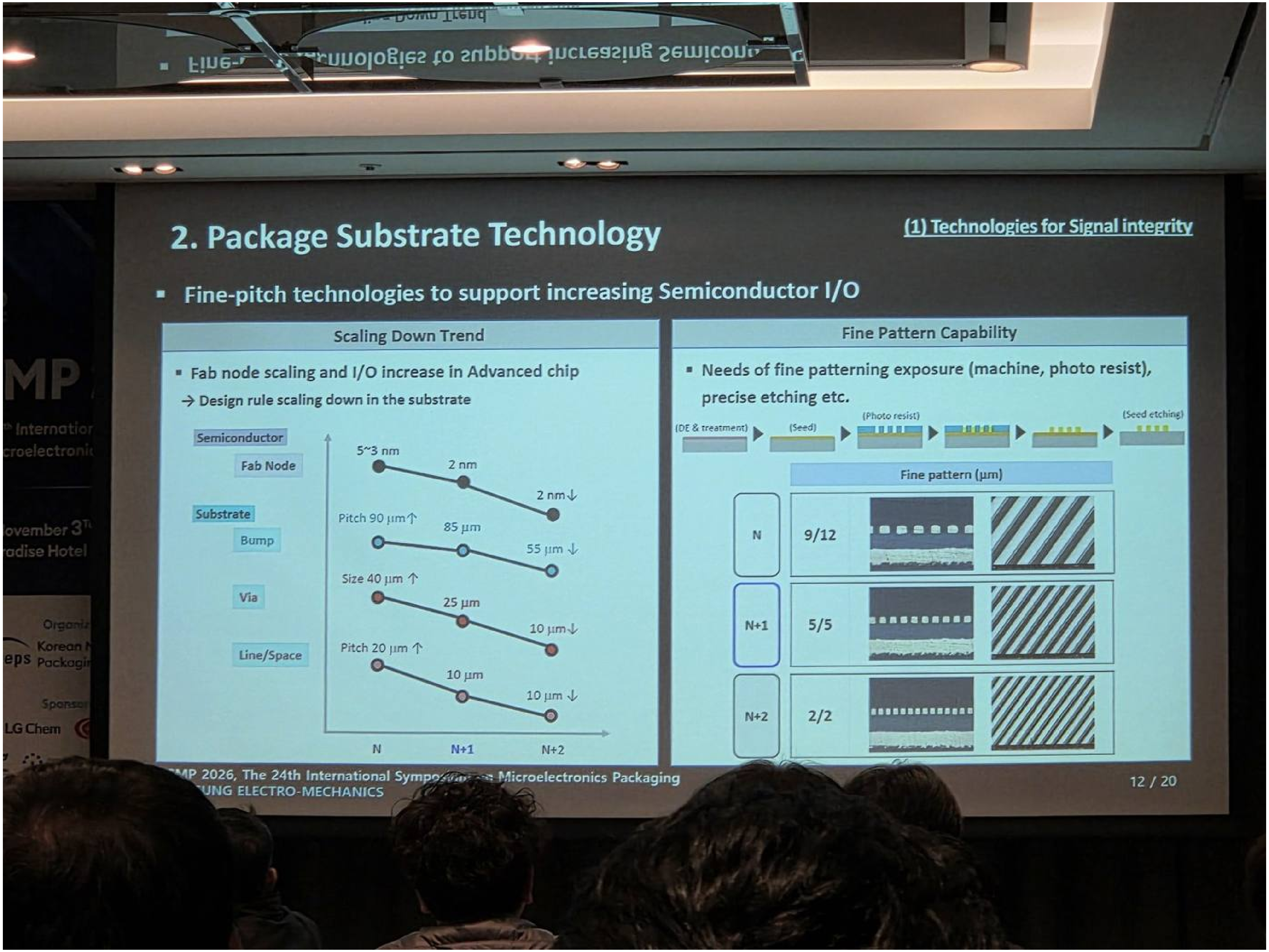
T12 고성능 기판의 신호 전력 및 기계적 설계

Sanghun Kim · Samsung Electro-Mechanics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2. Package Substrate Technology

(1) Technologies for Signal integrity

- Fine-pitch technologies to support increasing Semiconductor I/O

Scaling Down Trend

- Fab node scaling and I/O increase in Advanced chip

→ Design rule scaling down in the substrate

Semiconductor
Fab Node
5~3 nm
2 nm
2 nm↓
Substrate
Bump
Pitch 90 μm↑
85 μm
55 μm ↓
Via
Size 40 μm ↑
25 μm
10 μm↓
Line/Space
Pitch 20 μm ↑
10 μm
10 μm ↓
N
N+1
N+2

Fine Pattern Capability

- Needs of fine patterning exposure (machine, photo resist), precise etching etc.

(DE & treatment)
(Seed)
(Photo resist)
(Seed etching)

	Fine pattern (μm)
N	9/12
N+1	5/5
N+2	2/2

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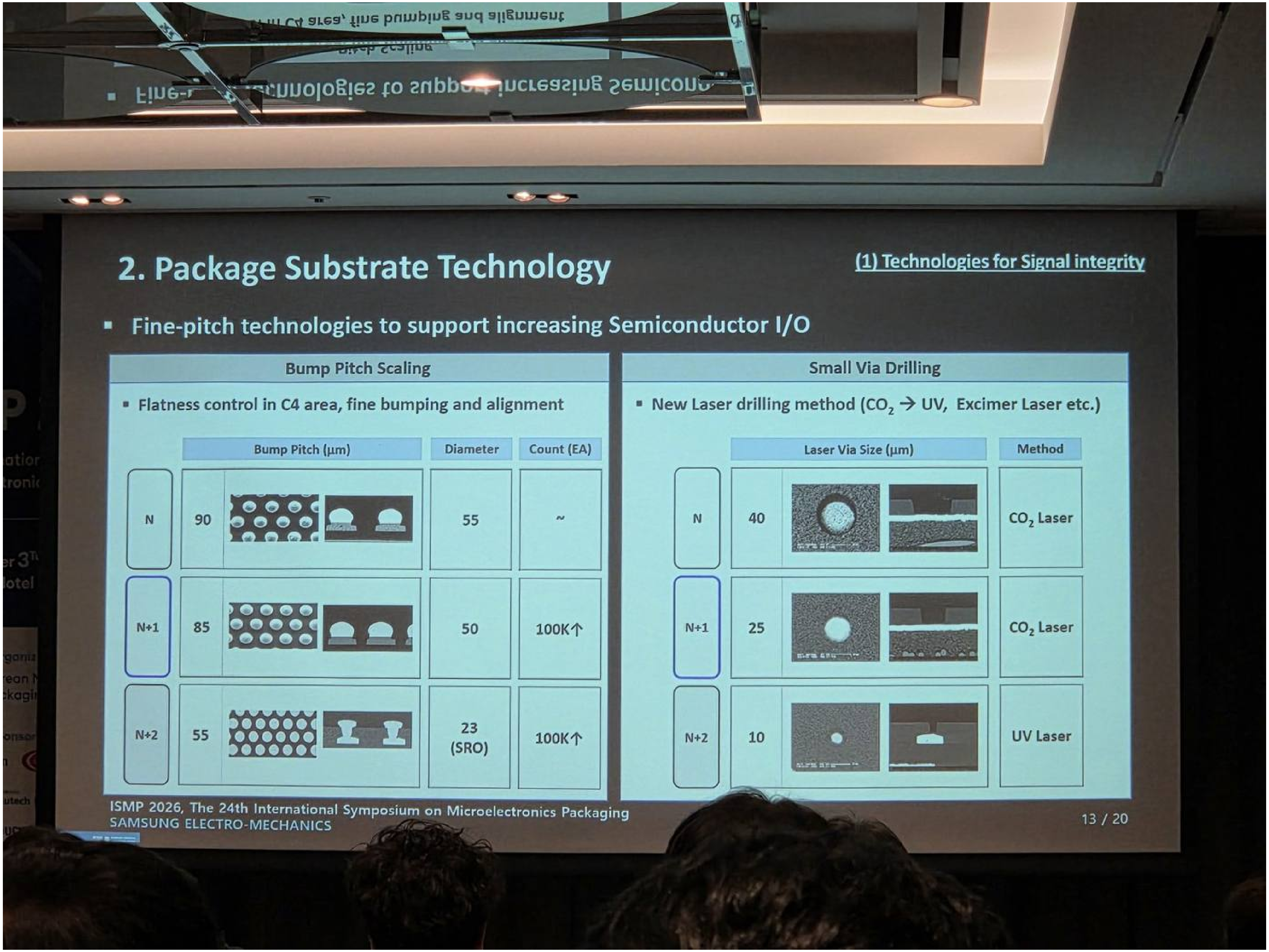

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(1) Technologies for Signal integrity

- Fine-pitch technologies to support increasing Semiconductor I/O

Bump Pitch Scaling

- Flatness control in C4 area, fine bumping and alignment

	Bump Pitch (µm)	Diameter	Count (EA)
N	90	55	~
N+1	85	50	100K↑
N+2	55	23 (SRO)	100K↑

Small Via Drilling

- New Laser drilling method (CO_2 → UV, Excimer Laser etc.)

	Laser Via Size (µm)	Method
N	40	CO_2 Laser
N+1	25	CO_2 Laser
N+2	10	UV Laser

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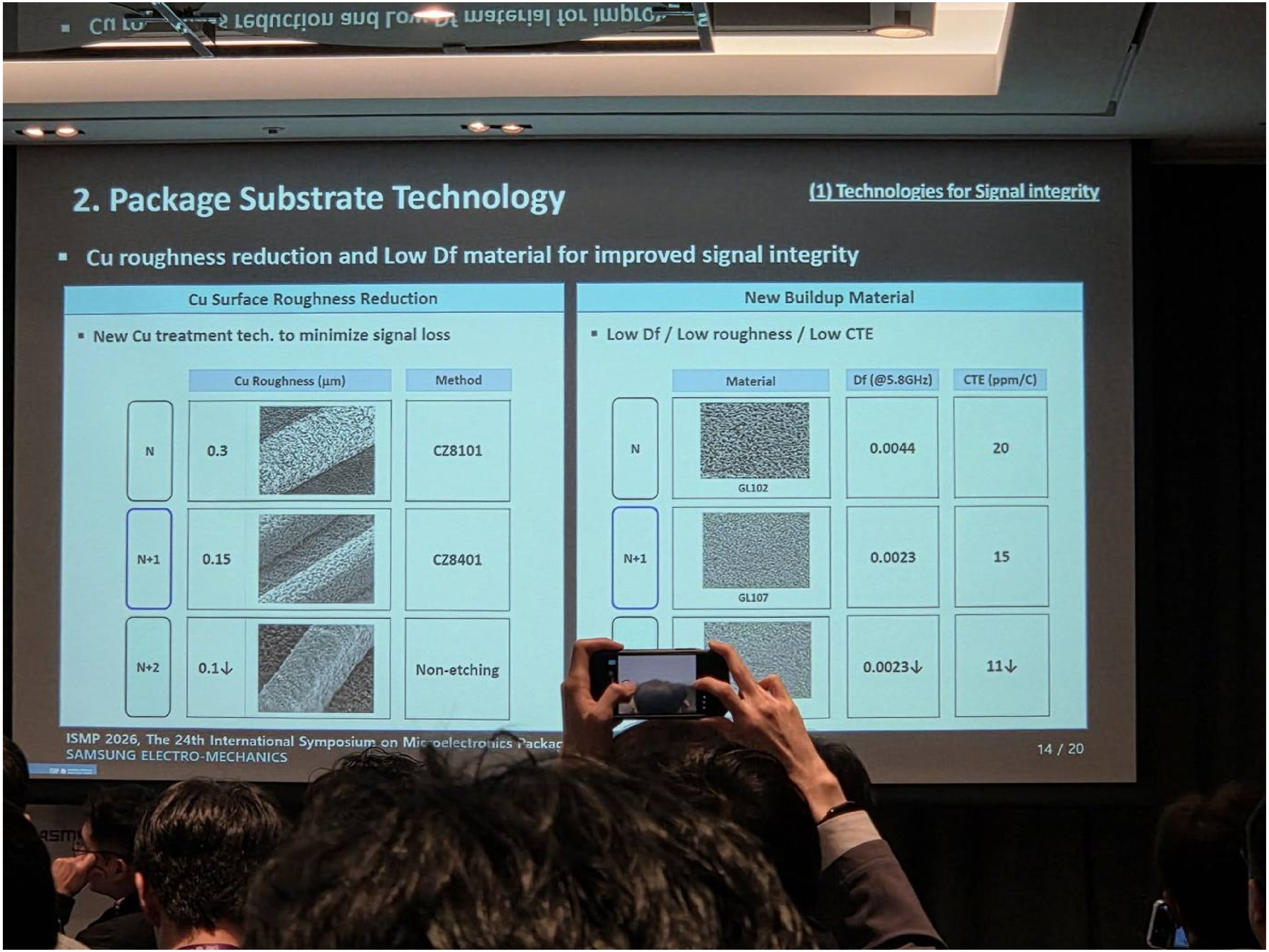

2. Package Substrate Technology

(1) Technologies for Signal integrity

- Cu roughness reduction and Low Df material for improved signal integrity

Cu Surface Roughness Reduction

- New Cu treatment tech. to minimize signal loss

	Cu Roughness (μm)	Method
N	0.3	CZ8101
N+1	0.15	CZ8401
N+2	0.1↓	Non-etching

New Buildup Material

- Low Df / Low roughness / Low CTE

	Material	Df (@5.8GHz)	CTE (ppm/C)
N	GL102	0.0044	20
N+1	GL107	0.0023	15
		0.0023↓	11↓

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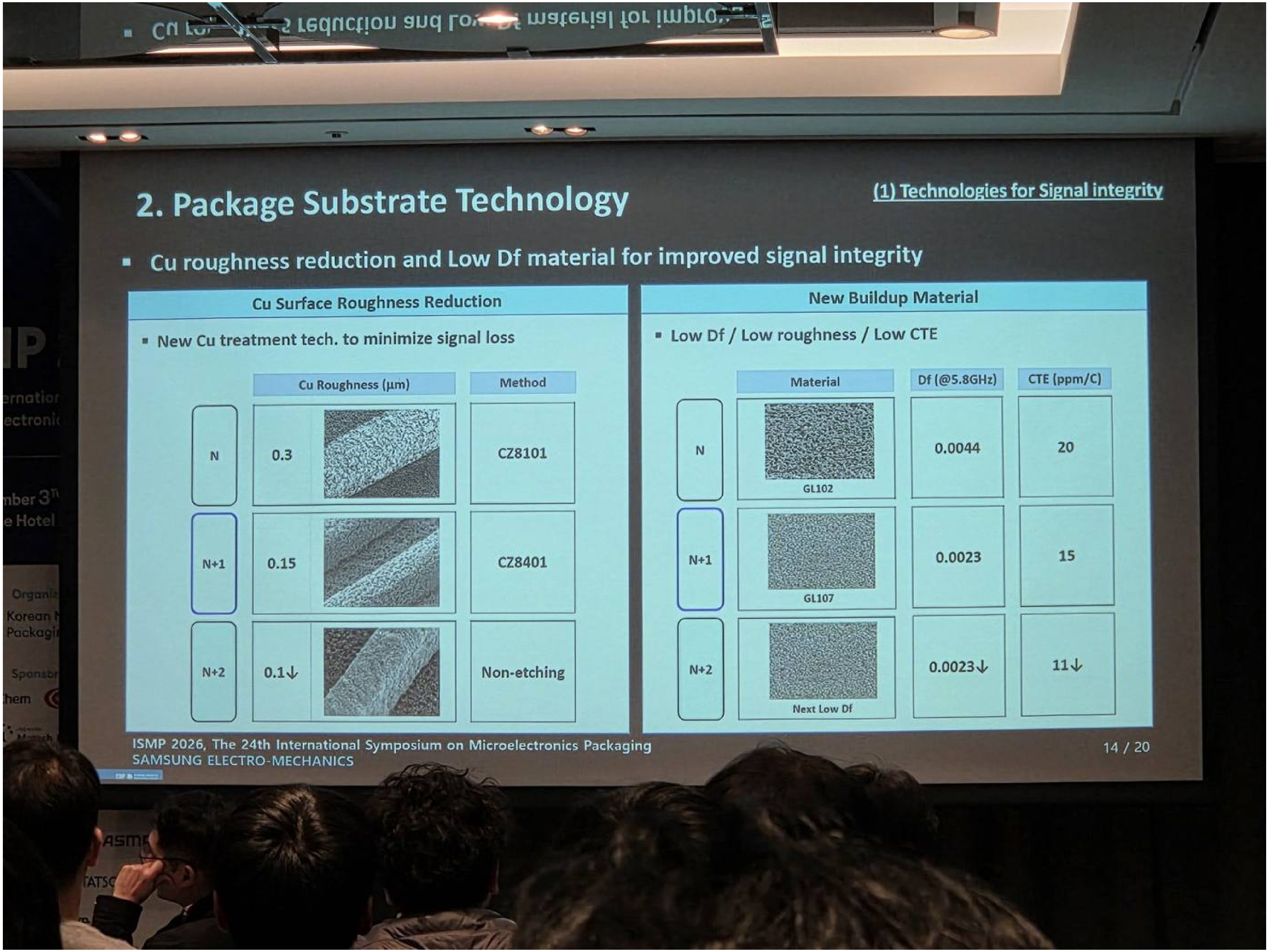

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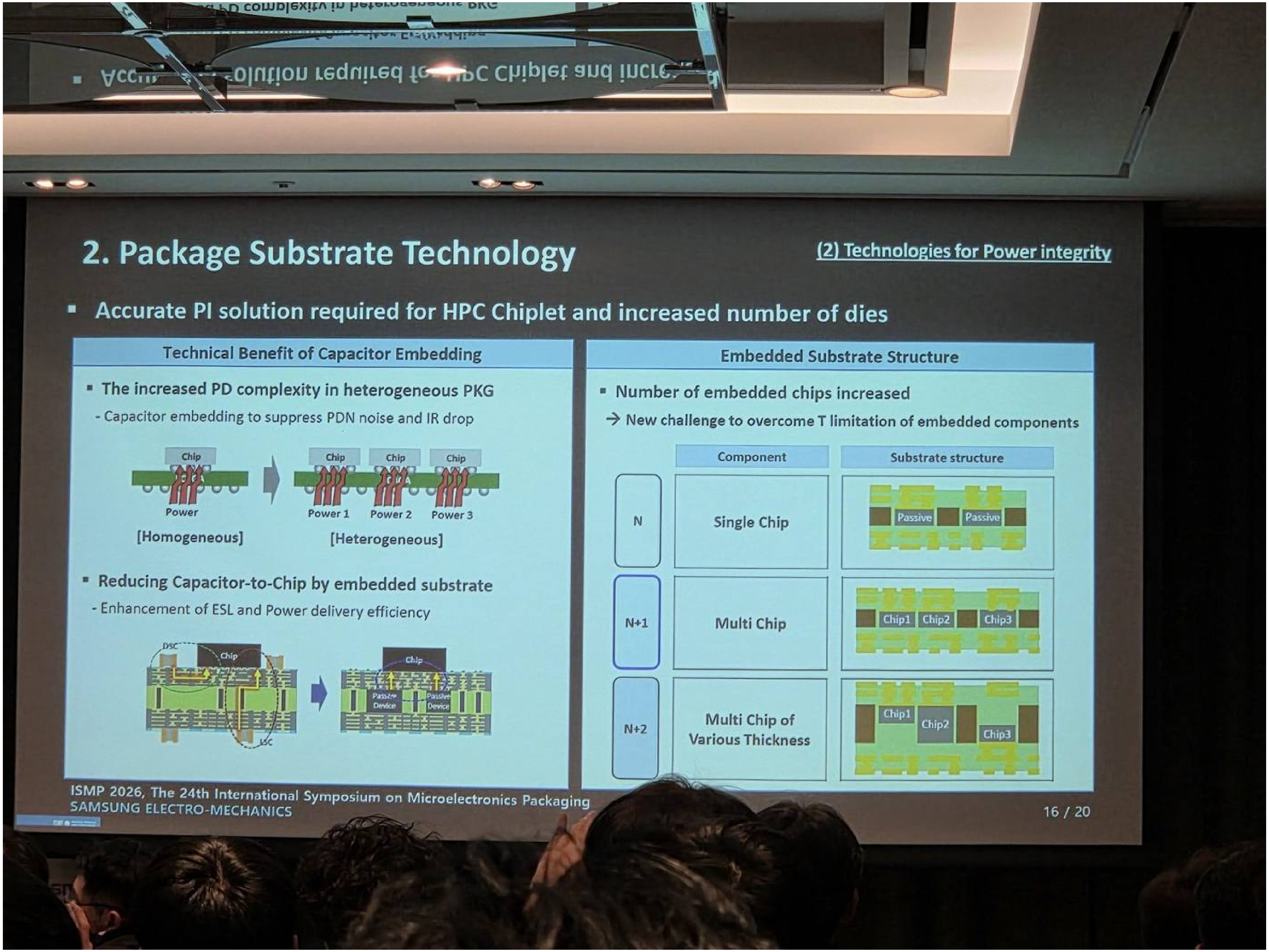

2. Package Substrate Technology

(2) Technologies for Power integrity

- Accurate PI solution required for HPC Chiplet and increased number of dies

Technical Benefit of Capacitor Embedding

- The increased PD complexity in heterogeneous PKG
  - Capacitor embedding to suppress PDN noise and IR drop

Chip
Power
[Homogeneous]
Chip
Chip
Chip
Power 1
Power 2
Power 3
[Heterogeneous]

- Reducing Capacitor-to-Chip by embedded substrate
  - Enhancement of ESL and Power delivery efficiency

DSC
Chip
LSC
Chip
Passive Device
Passive Device

Embedded Substrate Structure

- Number of embedded chips increased
  → New challenge to overcome T limitation of embedded components

	Component	Substrate structure
N	Single Chip	Passive Passive
N+1	Multi Chip	Chip1 Chip2 Chip3
N+2	Multi Chip of Various Thickness	Chip1 Chip2 Chip3

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2. Package Substrate Technology

(3) Technologies for Mechanical integrity

Large area/layer

Hybrid Build-up

Glass Core

[Signal integrity] Scaling Down (5)

[Signal integrity] Signal Performance (2)

[Power integrity] (1)

[Mechanical integrity] (3)

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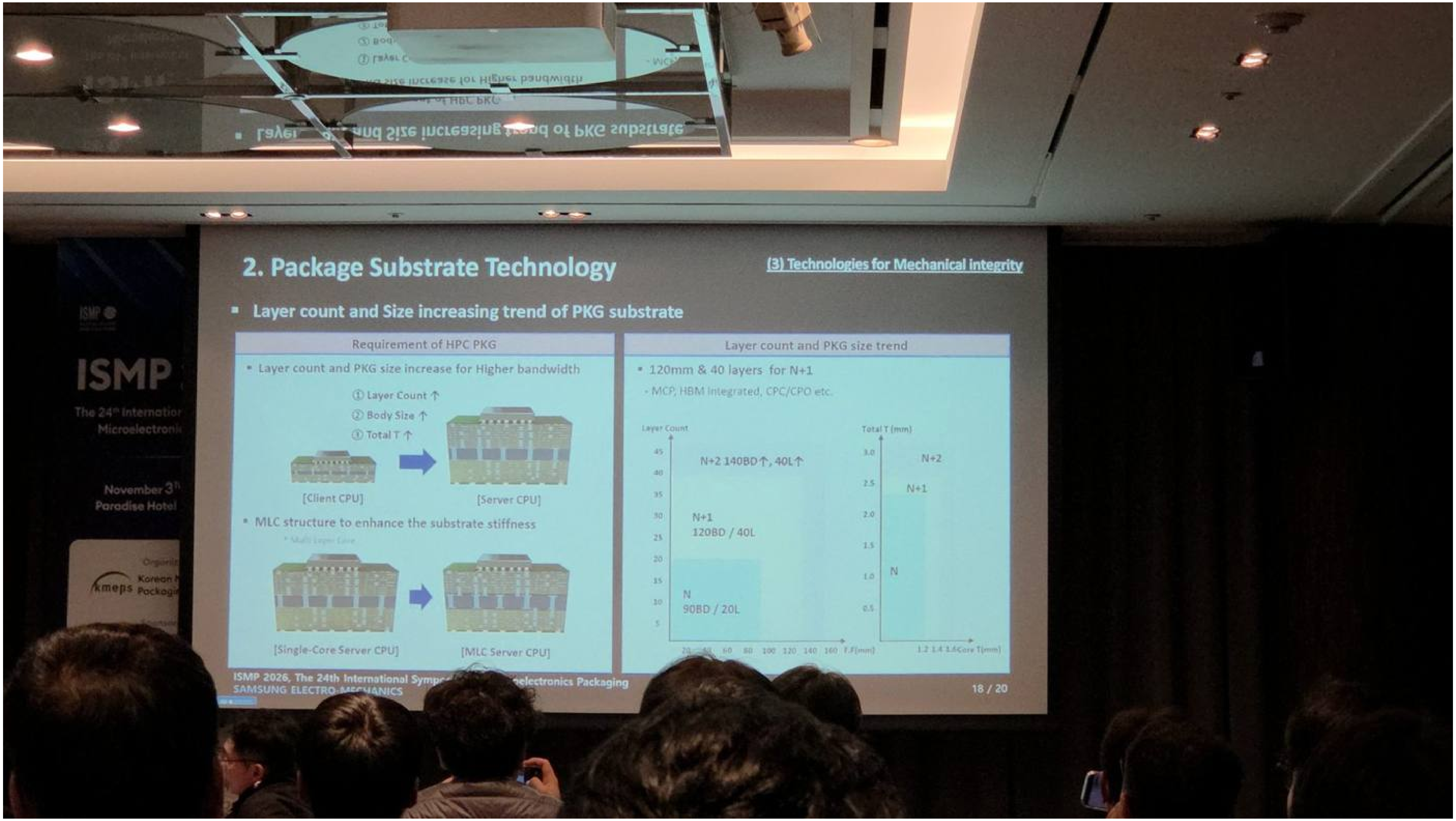

2. Package Substrate Technology

(3) Technologies for Mechanical integrity

- Layer count and Size increasing trend of PKG substrate

Requirement of HPC PKG

- Layer count and PKG size increase for Higher bandwidth

① Layer Count ↑
② Body Size ↑
③ Total T ↑

[Client CPU]
[Server CPU]

- MLC structure to enhance the substrate stiffness

* Multi Layer Core

[Single-Core Server CPU]
[MLC Server CPU]

Layer count and PKG size trend

- 120mm & 40 layers for N+1
  - MCP, HBM Integrated, CPC/CPO etc.

Layer Count
45
40
35
30
25
20
15
10
5
N+2 140BD↑, 40L↑
N+1
120BD / 40L
N
90BD / 20L
20
40
60
80
100
120
140
160
F.F(mm)
Total T (mm)
3.0
2.5
2.0
1.5
1.0
0.5
N+2
N+1
N
1.2 1.4 1.6Core T(mm)
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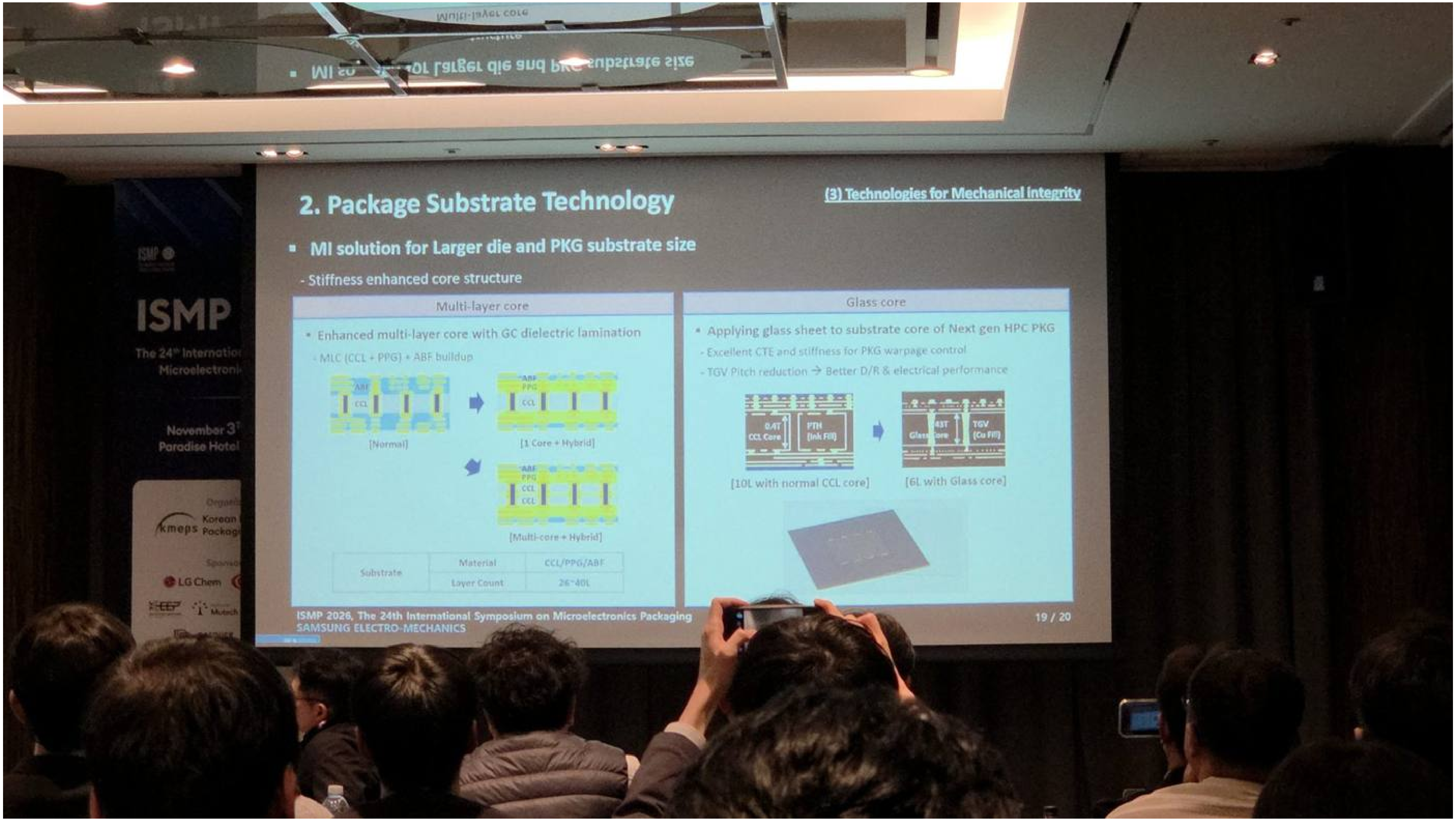
ABF
CCL
PPG
CCL
0.4T
CCL Core
PTH
(Ink Fill)
Glass Core
TGV
(Cu Fill)

2. Package Substrate Technology

(3) Technologies for Mechanical Integrity

- MI solution for Larger die and PKG substrate size

- Stiffness enhanced core structure

Multi-layer core

- Enhanced multi-layer core with GC dielectric lamination
  - MLC (CCL + PPG) + ABF buildup

[Normal]

[1 Core + Hybrid]

[Multi-core + Hybrid]

Substrate	Material	CCL/PPG/ABF
	Layer Count	26~40L

Glass core

- Applying glass sheet to substrate core of Next gen HPC PKG
  - Excellent CTE and stiffness for PKG warpage control
  - TGV Pitch reduction → Better D/R & electrical performance

[10L with normal CCL core]

[6L with Glass core]

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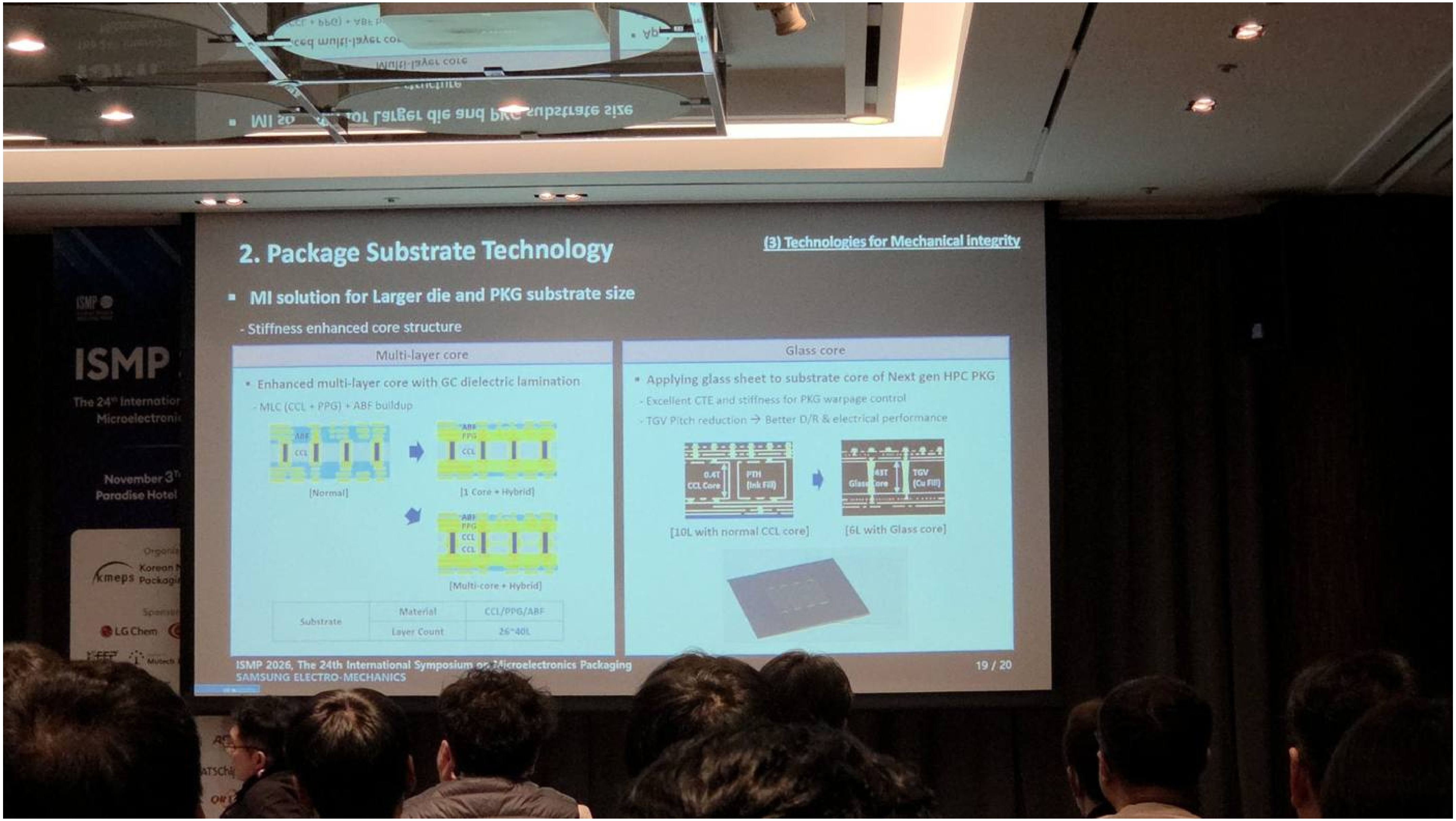

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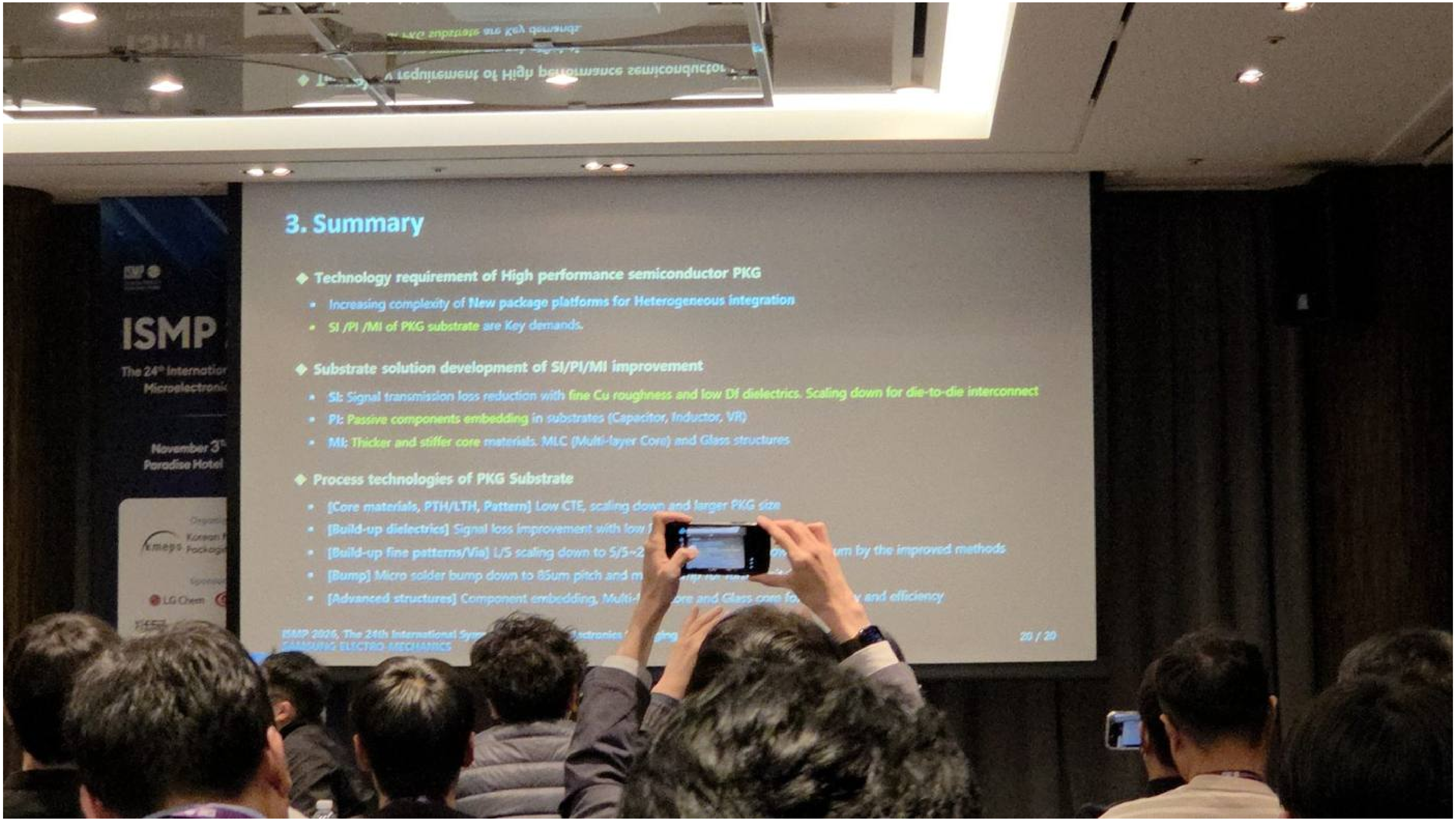
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3. Summary

◆ Technology requirement of High performance semiconductor PKG

- Increasing complexity of New package platforms for Heterogeneous integration
- SI /PI /MI of PKG substrate are Key demands.

◆ Substrate solution development of SI/PI/MI improvement

- SI: Signal transmission loss reduction with fine Cu roughness and low Df dielectrics. Scaling down for die-to-die interconnect
- PI: Passive components embedding in substrates (Capacitor, Inductor, VR)
- MI: Thicker and stiffer core materials. MLC (Multi-layer Core) and Glass structures

◆ Process technologies of PKG Substrate

- [Core materials, PTH/LTH, Pattern] Low CTE, scaling down and larger PKG size
- [Build-up dielectrics] Signal loss improvement with low [illegible]
- [Build-up fine patterns/Via] L/S scaling down to S/S~2 [illegible] um by the improved methods
- [Bump] Micro solder bump down to 85um pitch and m [illegible]
- [Advanced structures] Component embedding, Multi-[illegible] core and Glass core fo[illegible] and efficiency

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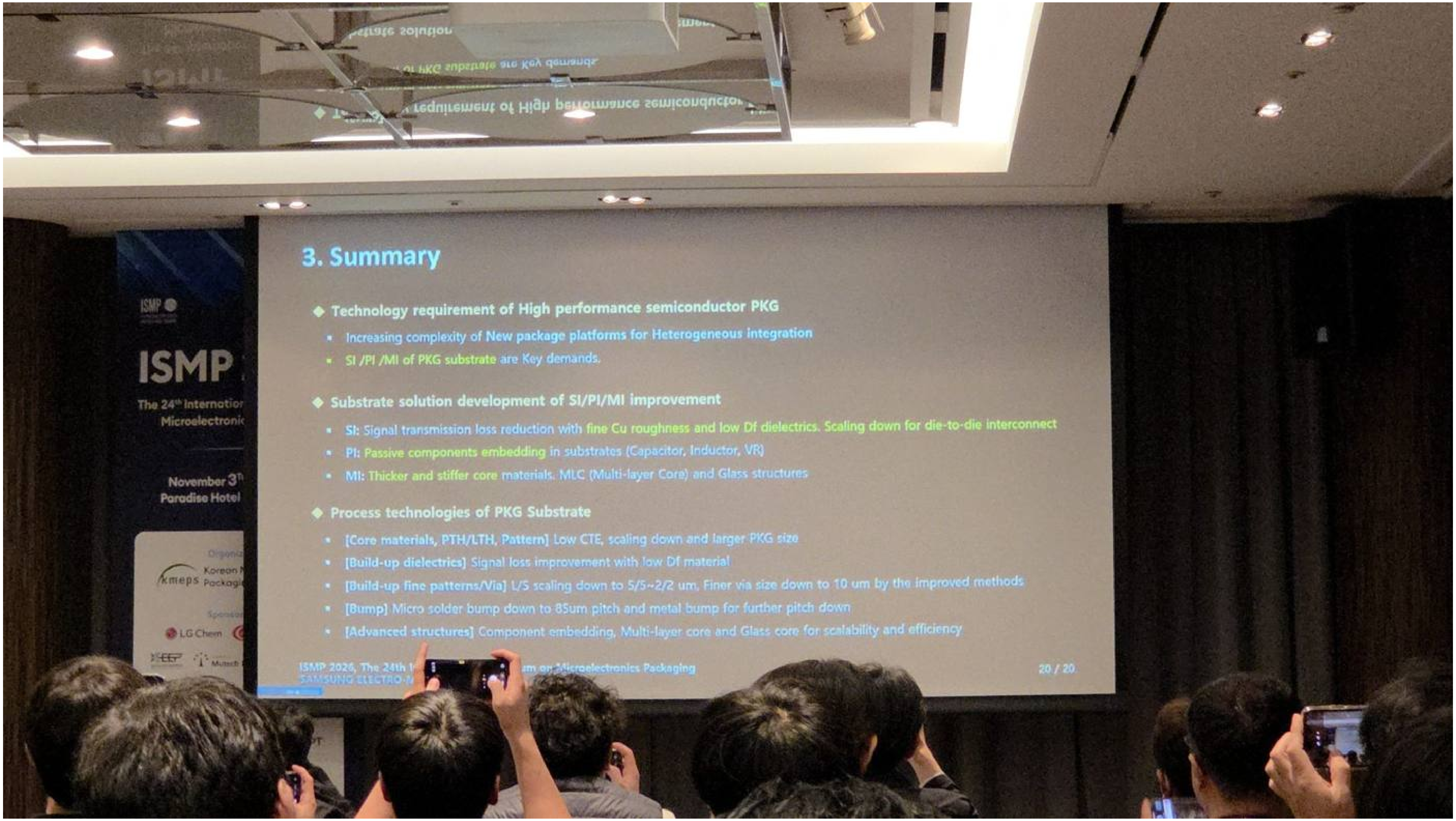

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- [Bump] Micro solder bump down to 85um pitch and metal bump for further pitch down
- [Advanced structures] Component embedding, Multi-layer core and Glass core for scalability and efficiency

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